Shayan Amouzadeh

Semnan, Iran | +98 919 332 3537 | shayanamz.7687@gmail.com
LinkedIn: linkedin.com/in/shayan-amouzadeh | GitHub: github.com/Shayan-Amz
Portfolio: shayan-amz.github.io | ORCID: orcid.org/0009-0002-5518-1729

RESEARCH INTERESTS

- **AI & Machine Learning:** Large Language Models (LLMs), Natural Language Processing (NLP), multimodal speech-to-text pipelines, and information retrieval.
- **Software Engineering & Computer Systems:** Automated testing and formal verification, high-performance and deterministic computing, computer graphics, and secure web architectures.

EDUCATION

B.Sc. in Computer Engineering (Software Engineering Specialization) | Sep 2022 – Sep 2026

Semnan University, Faculty of Electrical and Computer Engineering | Semnan, Iran

Academic record: Cumulative 15.77/20 | Last 2 years 16.93/20 | Final year 17.03/20 | Selected core CS/CE average 18.03/20 (21 courses). Grades on a 20-point scale.

Selected coursework (grade / 20): Information Technology (20.00), Programming Languages Design (19.50), Computer Networks Lab (19.50), Software Engineering (19.25), Operating Systems Lab (19.25), Web Programming (18.50), Linear Algebra (18.00), Data Structures & Algorithms (17.50).

UNDERGRADUATE THESIS & CAPSTONE PROJECT

Developer & System Architect — Intelligent Real Estate CRM Platform

B.Sc. Thesis: Design & Implementation of an Intelligent Real Estate Management System | Grade: 19.50/20 | Advisor: Dr. Morteza Dorrigiv, Semnan University | B.Sc. Capstone Project

Live project: amlak-e-man.ir | GitHub: Real-Estate-File-Manager-Project

- **Multi-tenant architecture & RBAC:** Architected and deployed a production CRM using PHP/PDO, a 7-table MySQL schema, 17 REST action handlers, and a zero-dependency vanilla-JavaScript PWA. Implemented version-gated runtime migrations, composite indexing, a 15-second change-token synchronization protocol, and three-role RBAC across three commercial tiers.
- **Voice and LLM extraction:** Built a human-reviewed Persian voice-to-form pipeline: browser-recorded 16 kHz mono PCM-16 WAV, provider-switchable Whisper-large-v3 speech recognition (Hugging Face, Groq, OpenAI), and OpenRouter LLM extraction. Added a 16-rule Persian JSON contract, prompt-injection guardrails, a 26-field server allow-list, and client-side spoken-number/currency normalization.
- **Matching, security & evaluation:** Implemented O(P) demand/property matching with normalized bidirectional city matching, ±15% area tolerance, three-mode budget constraints, and lazy-loaded Neshan Leaflet geocoding/deep links. Added 30-hour tenant-bound HMAC-SHA256 tokens with constant-time verification, per-request consultant revocation, SHA-256-masked guest-response tenant IDs, a 150 requests/minute sliding-window limiter, dual-stage image watermarking/sanitization, 47 tests, and an offline Persian WER/F1 console secured by HS256 JWTs (RFC 7519).

SELECTED TECHNICAL PROJECTS

IEEE 754 Single-Precision Floating-Point Unit

C++17 · CMake · Hardware Arithmetic Verification | GitHub

- **Implementation:** Built a bit-exact, compile-time (constexpr) C++17 binary32 model for addition, subtraction, multiplication, and division, including all four IEEE 754 rounding modes, subnormals, signed zeros, infinities, NaN propagation, and bitwise trace inspection.
- **Verification:** Differential-tested against the host FPU under fesetround(): 32,026,992 operand-pair checks with zero mismatches, enforced on GCC/Clang/AppleClang/MSVC CI. Improved an initial course adder from 58.2% bit-exact (1.4M samples) to 100% using sticky-bit alignment, round-to-nearest-even, and gradual underflow.

Deterministic Graphics Pipeline & CPU Software Rasterizer

C++17 · C99 · OpenGL 1.1 · FreeGLUT · Computer Graphics | GitHub

- **Architecture & rasterization:** Refactored an animated OpenGL scene into decoupled AnimationState and pure render() components. Authored softgl, a C99 deterministic rasterizer with 49 GL/GLUT entry points, Pineda half-space edge functions, pixel-center sampling, top-left fill, DDA lines, and Gouraud shading; verified with 28 self-test assertions.
- **Cross-platform testing:** Enforced pixel-exact CI across four scenes and GCC, AppleClang, and MSVC (up to 7,017 triangles and 1.26M fragments/frame at 1300×650). Resolved an ARM64 FMA-contraction discrepancy and removed 605 legacy OpenGL errors per frame.

SELECTED TECHNICAL PROJECTS (CONTINUED)

Bare-Metal ARMv7-A VGA Framebuffer Renderer

ARMv7-A Assembly · Python · Unicorn CPU Emulation | GitHub

- **Renderer & validation:** Wrote a bare-metal ARMv7-A renderer for the DE1-SoC 320×240 RGB565 memory-mapped VGA framebuffer (0xC8000000), using an AAPCS-compliant leaf routine and paired 16-bit pixel stores. Instrumented GNU and Zig builds with Python Unicorn in CI; verified exact 80/80/80 stripe geometry, clean halt, and zero out-of-bounds stores.
- **Optimization:** Reduced executed instructions from 353,286 to 116,201 (−67%; 3.04×), eliminating 9,216 out-of-bounds writes and 46,080 redundant padding writes.

End-to-End Box-Office Scraping & Content Recommender Pipeline

Python 3.10+ · Pandas · BeautifulSoup4 · Streamlit · Data Engineering | GitHub

- **Data pipeline:** Built a pip-installable src-layout pipeline for 2013–2024 (1392–1403 SH) Iranian box-office archives, producing a cleaned 686-film, 52-column dataset with outlier filtering, Persian numeral normalization, and 43 multi-label genre features (102 unique signatures).
- **Recommendation & quality:** Implemented deterministic exact-signature Top-K rankings by critic rating and box-office revenue in a cached Streamlit interface; added 14 pytest tests and Ruff CI across Python 3.10 and 3.12.

Exact-Optimal Fractional Knapsack & DP Benchmark Suite

C++17 · Python Benchmarking · Algorithmic Optimization | GitHub

- **Algorithms:** Implemented a header-only O(n log n) C++17 fractional-knapsack solver with a division-free cross-product comparator and deterministic tie-breaking; paired it with an O(nW) rolling-array 0/1 DP reference, an LP-relaxation bound for branch-and-bound, and a formal exchange-argument proof.
- **Evaluation:** Validated 328,582 automated checks with zero failures across 20,000 randomized instances. Benchmarked 1.86 ms versus 26.38 s for the O(nW) reference at n=20,000; measured integrality-gap decay from 30.0% (n=5) to 0.05% (n=200).

PROFESSIONAL EXPERIENCE

Computer Engineer & IT Specialist | Mar 2025 – Present (Full-Time)

Khakab Sazeh Ghomes Co. | Semnan, Iran

- **IT operations:** Manage internal networks, computers, user accounts, hardware/software provisioning, installation, troubleshooting, and day-to-day system administration.
- **Automation & reporting:** Build relational-database and reporting workflows with Microsoft Access and Excel; automate financial/engineering billing, project scheduling, resource tracking, and recurring office/documentation tasks using Excel, Access, and Microsoft Project.

Game Developer & Software Engineering Intern | Jul 2025 – Dec 2025

Sargarmi Sazan Aseman Omid (Game Development Studio) | Iran

- **Gameplay systems:** Developed core Unity/C# gameplay mechanics using the Entitas Entity Component System (ECS), separating component state from reactive execution systems on proprietary studio projects.
- **UI & engineering:** Built uGUI and TextMeshPro interfaces, contributed profiling and optimization, and developed an open-source Unity 6 uGUI/URP 2D directory with LINQ-powered search and sorting. Collaborated with Git and Plastic SCM (Unity Version Control).

TECHNICAL SKILLS

Programming languages: C++17, C, Python 3, ARMv7-A Assembly, C#, PHP 8, JavaScript, SQL, HTML5/CSS3
AI, NLP & data: LLM structured extraction, Whisper speech-to-text, Persian NLP normalization, Pandas, BeautifulSoup4, Streamlit, WER/F1 evaluation
Systems & verification: IEEE 754 arithmetic, differential FPU testing (fesetround), deterministic rasterization, Unicorn CPU emulation, memory-mapped I/O
Web, databases & security: REST-style APIs, PWA/SPA, Neshan Leaflet SDK, MySQL/MariaDB/SQLite (PDO, migrations), HMAC-SHA256, HS256 JWT (RFC 7519), rate limiting
Graphics, engines & tools: OpenGL/FreeGLUT, Unity 6 (uGUI, URP 2D, LINQ), Entitas ECS, GitHub Actions, CMake, GCC/Clang/MSVC, Docker, Pytest, Ruff, Puppeteer

LANGUAGES

Persian: Native | **English:** Advanced working proficiency